OR-AS Operations Research Applications and Solutions	**Case Name:** Water Intake and Canal Rehabilitation **Project Owner:** Eder Omar Vilca Carrillo	Sector	Construction
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Processed by	Amin Azimi		User defined distributions
Date	2025	Project Control	Automatic tracking
File Name	C2025-01		Tracking based on user input

1. Project description

Project authenticity

The rehabilitation of a water intake structure (bocatoma), desander, and irrigation canal to restore and enhance water flow efficiency.
The project consists of activity and cost data that were obtained directly from the actual project owner

2. Project properties

2.1. Baseline Schedule

General	
# Activities	45
Planned Duration (PD)	57 days
Budget At Completion (BAC)	278,057.33€
Renewable Resources	5 Types
Consumable Resources	0 Types

Network topology	
Serial/Parallel (SP)	36%
Activity Distribution (AD)	41%
Length of Arcs (LA)	0%
Topological Float (TF)	25%

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	3.068523	13.57468	5.012909
CRI-rho	47.33227	12.81577	-2.55436
CRI-tau	91.04826	26.46582	-2.81044

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	34.64444	43.8749	0.702286
SI	46.83245	42.18012	0.256011
SSI	7.014305	13.92701	3.585681
CRI-r	10.02051	11.98819	3.397331
CRI-rho	10.05055	11.81109	3.617428
CRI-tau	7.45679	8.231369	3.579023

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	40.09	6.95	-5.78
CRI-rho	40.09	6.95	-5.78
CRI-tau	40.09	6.95	-5.78

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	15.74	12.20
PV - SPI	19.13	13.10
PV - SCI	19.14	13.10
ED - 1	13.81	10.33
ED - SPI	17.37	11.33
ED - SCI	17.38	11.33
ES - 1	12.84	9.80
ES - SPI(t)	15.32	10.73
ES - SCI(t)	15.34	10.74

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0	0
CPI	0.13	0
SPI	6.00	0.59
SPI(t)	5.27	0.43
SCI	6.04	0.59
SCI(t)	5.31	0.43
0.8 CPI + 0.2 SPI	1.55	0.73
0.8 CPI + 0.2 SPI(t)	1.18	0.38

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 4 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	2108.50	-66768.44	-106	1.04	0.48	0.47	0.83
std dev	1249.28	49729.09	95.19	0.03	0.36	0.32	0.20
final	3936.43	0	-24	1.01	1	0.95	1

2.3.3.2. Time forecasting

PD	57 Days

Real Duration	62 Days

Delay	8%

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	73.5	10.87	20.97	-18.54
PV - SPI	178.5	109.31	190.32	-187.90
PV - SCI	169.25	97.76	175.40	-172.98
ED - 1	71.75	9.74	15.72	-15.72
ED - SPI	179.25	108.23	189.12	-189.11
ED - SCI	170.75	97.04	175.40	-175.40
ES - 1	72.25	11.90	16.53	-16.53
ES - SPI(t)	164	80.57	164.51	-164.51
ES - SCI(t)	158.25	78.25	155.24	-155.24

2.3.3.3. Cost forecasting

BAC	278,057.33€

Real Cost	278,057.33€

On Budget	0%

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	279885.27	1249.28	0.65	-0.65
CPI	271051.86	8770.73	2.51	2.51
SPI	772834.62	494528.84	177.94	-177.94
SPI(t)	686323.89	301911.91	146.82	-146.82
SCI	733520.68	440678.29	163.80	-163.80
SCI(t)	659330.52	288173.67	137.12	-137.12
0.8 CPI + 0.2 SPI	299527.32	14326.19	7.72	-7.72
0.8 CPI + 0.2 SPI(t)	298775.58	15539.71	7.45	-7.45